EFY TESTED
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DEVICE TO IDENTIFY And Classify Objects

SOMNATH BERA

This device can recognise items like vegetables and fruits and help in segregating them as well as classify them as per their size and other features. It uses the edge Impulse ML tool that can be deployed on various platforms and boards, such as Arduino, ESP32 camera, ESPEye, laptop PC, cellphone, Raspberry Pi computer, and more. You can create the output in the form of C++ library, which can be used almost anywhere.

Let us say you need to segregate vegetables like lemons, onions, and tomatoes, or pens and pencils. You need just a Raspberry Pi board or ESP32 camera and a few relays to segregate them. Here, we shall use an ESP32 cam to identify vegetables, as an example.

When the cam detects a tomato, onion, or lemon, for instance, the relays will get actuated to open the basket containing the vegetables. The author's prototype is shown in Fig. 1. The components needed for the project are mentioned in the table for Bill of Material.

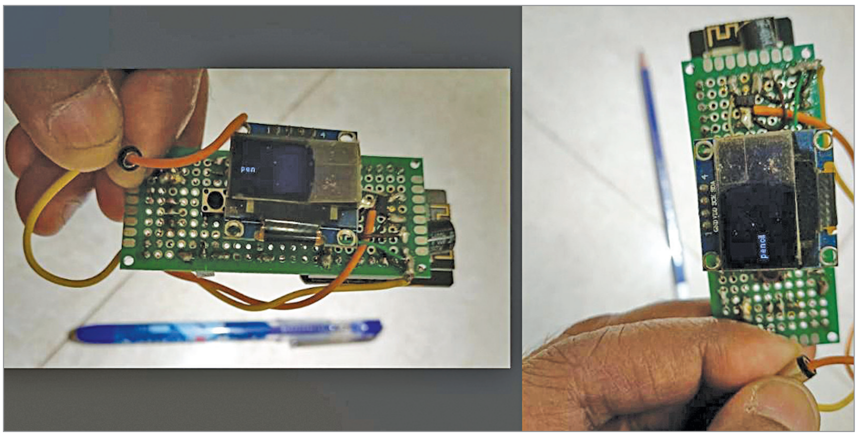

Fig. 1: Author's prototype

Preparing the ML model

To start the project, open edgeimpulse.com and create a new project. You have to collect photographs of the items to be segregated in groups and as single pieces from several angles for the device to recognise them correctly, and edgeimpulse

BILL OF MATERIAL

Components	Quantity
ESP32 cam/ESP32 TTT GO (MOD1)	1
LM117 voltage regulator (MOD3)	1
5V with relay	1
BC547 transistor (T1)	1
SSD1306 OLED (MOD2)	1
100μF capacitor (C1)	2
Diode (D1)	1
FTDI USB programmer	1

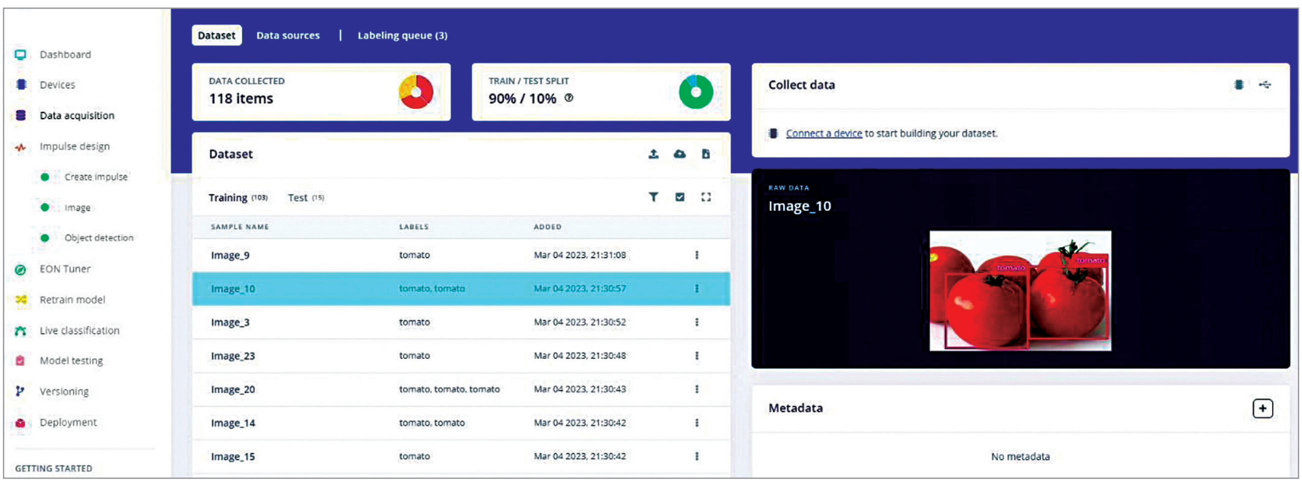

Fig. 2: Collecting dataset